

Adaptive Heuristic Portfolio TSP Report

Overview

- Condition count: 8.
- Best held-out TSPLIB gap: phase8_llm_evolved_adaptive_controller.
- Best combined transfer gap: phase8_llm_evolved_adaptive_controller.
- Lowest selector regret: phase8_llm_evolved_adaptive_controller.
- Pareto-efficient conditions: phase8_full_solver_evolution, phase8_llm_evolved_adaptive_controller, phase8_random_portfolio, phase8_supervised_ml_selector.

Run Metadata

- run_name: run_20260520_224252_p
- started_at_local: 2026-05-20 22:42:52
- finished_at_local: 2026-05-20 22:59:07
- duration_hhmm: 00:16
- duration_seconds: 974.878
- seed_offset: 15000
- replicate_label: p
- judge_status: enabled
- judge_provider: openai
- judge_model: gpt-4.1-mini

Condition Comparison

Condition	Mode	Replay	Final TSPLIB Gap	Selector Regret	Run time (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_best_single_fixed_heuristic	best_fixed	none	0.07226	0.0	699.734114	0.07226	0.007284	0.03122	0.0	0.0	False
phase8_random_portfolio	random_portfolio	none	0.160733	0.088473	195.759071	0.160733	0.136539	0.145452	0.0	0.0	True
phase8_oracle_selector	oracle_selector	none	0.07226	0.0	699.788357	0.072266	8.9e-05	0.026678	0.0	0.0	False

Condition	Mode	Replay	Final TSPLI B Gap	Selector Regret	Runtime (ms)	Runtime -Adjusted Gap	Family Gap	Transfer Gap	Mean Novelty	Mean Complexity	Pareto Efficient
phase8_supervised_ml_selector	supervised_selector	none	0.07226	0.0	699.511157	0.07226	0.007284	0.03122	0.0	0.0	True
phase8_llm_static_selector	static_selector	none	0.216874	0.144614	165.082929	0.216874	0.024668	0.095481	0.0	1.4	False
phase8_llm_evolved_adaptive_controller	adaptive_controller	none	0.067008	-0.005252	2049.241129	0.19624	8.9e-05	0.024743	0.611081	0.62	True
phase8_llm_evolved_controller_diversity_failure_replay	adaptive_controller	diversity_failure	0.214311	0.142051	264.935986	0.214311	0.023657	0.093898	0.489403	0.62	False
phase8_full_solver_evolution	full_solver	none	0.213387	0.141127	99.585957	0.213387	0.150269	0.173523	0.92146	0.56	True

Condition Notes

phase8_best_single_fixed_heuristic

- Execution mode best_fixed with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 699.734114 ms, runtime inflation 0.0, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.
- Fixed heuristic choice: farthest_insertion_two_opt.

phase8_random_portfolio

- Execution mode random_portfolio with replay none.
- Final held-out gap 0.160733, selector regret 0.088473, runtime-adjusted gap 0.160733.
- Family transfer 0.136539 and combined transfer 0.145452.
- Runtime 195.759071 ms, runtime inflation -0.720238, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_oracle_selector

- Execution mode oracle_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.072266.

- Family transfer 8.9e-05 and combined transfer 0.026678.
- Runtime 699.788357 ms, runtime inflation 7.8e-05, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: False.

phase8_supervised_ml_selector

- Execution mode supervised_selector with replay none.
- Final held-out gap 0.07226, selector regret 0.0, runtime-adjusted gap 0.07226.
- Family transfer 0.007284 and combined transfer 0.031222.
- Runtime 699.511157 ms, runtime inflation -0.000319, mean novelty 0.0, and complexity 0.0.
- Pareto efficient: True.

phase8_llm_static_selector

- Execution mode static_selector with replay none.
- Final held-out gap 0.216874, selector regret 0.144614, runtime-adjusted gap 0.216874.
- Family transfer 0.024668 and combined transfer 0.095481.
- Runtime 165.082929 ms, runtime inflation -0.764078, mean novelty 0.0, and complexity 1.4.
- Pareto efficient: False.
- Controller signature: nearest_neighbor_multistart:cluster_first_local_search:farthest_insertion_two_opt:annealed_multi_start:stag3:cand2:restart1.
- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": 2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selector over frozen TSP heuristic portfolio with bounded schedule-tuning parameters; prefers bottleneck/cluster-aware methods only when descriptors justify it.", "failed_perturbation_threshold": 2, "grid_like_heuristic": "farthest_insertion_two_opt", "instance_adaptive_rules": [{"use": "grid_like_heuristic", "when": {"grid_likeness_gte": 0.6}}, {"use": "two_cluster_bottleneck_heuristic", "when": {"bottleneck_score_gte": 0.62}}, {"use": "clustered_heuristic", "when": {"clusteredness_score_gte": 0.42}}, {"use": "corridor_heuristic", "when": {"corridor_score_gte": 0.5}}, {"use": "nearest_neighbor_trap_heuristic", "when": {"nearest_neighbor_trap_score_gte": 0.26}}], "low_improvement_threshold": 0.08, "name": "tsp_frozen_portfolio_instance_adaptive_v1", "nearest_neighbor_trap_heuristic": "edge_preserving_restart", "random_like_heuristic": "nearest_neighbor_multistart", "restart_offset": 1, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 3, "temperature_scale": 1.15, "tie_breaker": {"deterministic_slope": {"key": "distance_cv", "prefer_lower": true}, "prefer": ["default_heuristic"]}, "time_budget_trigger": 0.45, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_adaptive_controller

- Execution mode adaptive_controller with replay none.
- Final held-out gap 0.067008, selector regret -0.005252, runtime-adjusted gap 0.19624.
- Family transfer 8.9e-05 and combined transfer 0.024743.
- Runtime 2049.241129 ms, runtime inflation 1.9286, mean novelty 0.611081, and complexity 0.62.
- Pareto efficient: True.
- Controller signature: annealed_multi_start:cluster_first_local_search:farthest_insertion_two_opt:farthest_insertion_two_opt:stag4:cand-2:restart2.

- Controller rule summary: {"acceptance_bias": 0.006, "candidate_limit_offset": -2, "clustered_heuristic": "cluster_first_local_search", "corridor_heuristic": "limited_three_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive HH controller over frozen TSP portfolio; selects structure-matched specialist heuristics and uses conservative stagnation/low-improvement retuning to avoid thrashing.", "failed_perturbation_threshold": 3, "grid_like_heuristic": "candidate_pruned_two_opt", "low_improvement_threshold": 0.12, "name": "tsp_hh_oracle_style_portfolio_controller_v4", "nearest_neighbor_trap_heuristic": "edge_preserving_restart", "random_like_heuristic": "annealed_multi_start", "restart_offset": 2, "stagnation_switch_heuristic": "farthest_insertion_two_opt", "stagnation_threshold": 4, "temperature_scale": 1.25, "time_budget_trigger": 0.7, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_llm_evolved_controller_diversity_failure_replay

- Execution mode adaptive_controller with replay diversity_failure.
- Final held-out gap 0.214311, selector regret 0.142051, runtime-adjusted gap 0.214311.
- Family transfer 0.023657 and combined transfer 0.093898.
- Runtime 264.935986 ms, runtime inflation -0.621376, mean novelty 0.489403, and complexity 0.62.
- Pareto efficient: False.
- Controller signature: annealed_multi_start:cheapest_insertion_two_opt:farthest_insertion_two_opt:annealed_multi_start:stag4:cand3:restart2.
- Controller rule summary: {"acceptance_bias": 0.008, "candidate_limit_offset": 3, "clustered_heuristic": "cheapest_insertion_two_opt", "corridor_heuristic": "candidate_pruned_two_opt", "default_heuristic": "farthest_insertion_two_opt", "description": "Deterministic instance-adaptive selector over a frozen TSP heuristic portfolio using interpretable structure scores (cluster/bottleneck/grid/corridor/trap) plus an online stagnation trigger to switch from mostly 2-opt refinement to restart/annealed diversifica", "failed_perturbation_threshold": 3, "grid_like_heuristic": "farthest_insertion_two_opt", "low_improvement_threshold": 0.08, "name": "tsp_heuristic_hyperheuristic_oracle_proxy_v2", "nearest_neighbor_trap_heuristic": "edge_preserving_restart", "random_like_heuristic": "annealed_multi_start", "restart_offset": 2, "stagnation_switch_heuristic": "annealed_multi_start", "stagnation_threshold": 4, "temperature_scale": 1.2, "time_budget_trigger": 0.6, "two_cluster_bottleneck_heuristic": "farthest_insertion_two_opt"}

phase8_full_solver_evolution

- Execution mode full_solver with replay none.
- Final held-out gap 0.213387, selector regret 0.141127, runtime-adjusted gap 0.213387.
- Family transfer 0.150269 and combined transfer 0.173523.
- Runtime 99.585957 ms, runtime inflation -0.85768, mean novelty 0.92146, and complexity 0.56.
- Pareto efficient: True.

Judge Appendix

Conservative Interpretation of TSP Adaptive Heuristic Portfolio Results

Metric	phase8_llm_evolved_adaptive_controller	phase8_supervised_ml_selector	phase8_best_single_fixed_heuristic	phase8_random_portfolio	phase8_full_solver_evolution
Held-out TSPLIB gap	**0.0670** (best, lower is better)	0.0723	0.0723	0.1607	0.2134

Metric	phase8_llm_evolved_adaptive_controller	phase8_supervised_ml_selector	phase8_best_single_fixed_heuristic	phase8_random_portfolio	phase8_full_solver_evolution
Selector Regret vs Oracle	-0.00525 (negative, meaning slightly better)	0.0	0.0	0.0885	0.1411
Runtime-Adjusted Gap	0.1962	0.0723	0.0723	0.1607	0.2134
Runtime Inflation	1.93 (substantial runtime increase)	-0.0003 (no inflation)	0.0	-0.72 (runtime reduced)	-0.86 (runtime reduced)
Transfer Gap Cross-Family	0.0247 (lowest)	0.0312	0.0312	0.1455	0.1735
Pareto Efficient	Yes	Yes	No	Yes	Yes
Adaptation Efficiency	0.0235 (modest adaptation)	0.0	0.0	0.0	-0.0347 (negative)

Key Points

- **Primary endpoint (Held-out TSPLIB gap):**

The **phase8_llm_evolved_adaptive_controller** achieves the best performance with a gap of 0.0670, outperforming the static supervised ML selector (0.0723) and best single heuristic baseline (0.0723). This indicates meaningful improvement in solution quality on unseen TSPLIB instances.

- **Selector regret vs oracle portfolio:**

The oracle selector is an upper bound; the evolved adaptive controller has slightly negative selector regret (-0.00525), effectively matching or slightly surpassing the oracle's performance (likely due to statistical variation). This suggests highly effective instance-adaptive control.

- **Runtime-adjusted gap / inflation:**

The evolved adaptive controller incurs a significant runtime inflation (~1.93x) compared to others. While it improves solution quality, this comes at a substantial runtime cost. The static ML selector achieves similar held-out gap with zero runtime inflation, offering a better runtime-quality tradeoff.

- **Pareto efficiency:**

The evolved adaptive controller, supervised ML selector, random portfolio, and full solver evolution are Pareto efficient. However, random portfolio and full solver evolution yield substantially worse solution quality.

- **Cross-family transfer:**

The evolved adaptive controller exhibits the best transfer gap (0.0247), indicating strong generalization across heterogeneous TSP families.

- **Adaptation efficiency:**

Evolved adaptive controller shows small positive adaptation efficiency (0.0235), indicating limited but present benefit from adaptation versus static selection. Static selectors have zero adaptation.

- ****Interpretability & control:****

The evolved adaptive controller uses interpretable, structure-based heuristic selection with conservative stagnation triggers, avoiding thrashing—a promising approach to hyper-heuristic control rather than discovering new heuristics.

Summary and Recommendations

- The ****phase8_llm_evolved_adaptive_controller**** shows the best held-out TSPLIB gap and selector regret near zero, demonstrating effective and interpretable instance-adaptive hyper-heuristic control that closely approaches oracle portfolio performance.
- However, this improvement comes with almost doubling average runtime, which may limit deployability unless runtime budgets are flexible.
- The ****phase8_supervised_ml_selector**** achieves competitive TSPLIB gap with no runtime inflation, representing a strong static adaptive baseline.
- The best single fixed heuristic already provides a reasonable baseline, but adaptive controllers outperform it in held-out generalization and transfer.
- Adaptive control with minimal runtime inflation remains a key goal; future work should explore runtime-quality trade-offs further.
- No new heuristics were discovered; all gains derive from improved adaptive control over a frozen heuristic portfolio, as intended.

****Conservative conclusion:**** The evolved adaptive controller effectively improves held-out TSP solution quality by leveraging interpretable instance features to select heuristics adaptively. It closes the gap with the oracle portfolio but at notable runtime cost. Static supervised selectors offer a compelling compromise with similar quality and better efficiency. The framework shows promising cross-family generalization and interpretable control, meriting further investigation to optimize runtime performance.